

OUTPUT D12 - Report - WORK PACKAGE WP3

Written Summary of the Research

Project Identification:

Project Title: A Statistical Approach to the Quantification and Monitoring of Neural Network Training

Project Acronym: SAMQNN

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1. Introduction

Quantisation is among the most effective techniques for compressing neural networks: converting weights from a 32-bit floating-point (FP32) representation to lower bit precision (typically 8 or 4 bits) substantially reduces both the memory and computational demands of the model, yet inevitably alters its weight distribution and may lead to a loss of accuracy.

The previous deliverable D10 demonstrated that comparing weight distributions can be integrated directly into the training of a quantised model with negligible overhead, and that the statistical index responds to the severity of quantisation. Deliverable D12 implements these findings into a functional training strategy, with emphasis on the methodological soundness of the quantisation.

A key difference from simplified implementations is that the strategy does not merely simulate quantisation combined with full-precision training — such an approach would allow the weights to drift off the quantisation grid back to arbitrary real values during training, so the resulting model would not in fact be quantised. Instead, genuine quantisation-aware training is implemented, in which the forward pass uses quantised weights and the backward pass employs a straight-through estimator; the genuineness of the quantisation is also verified numerically in the notebook.

2. Methodology

2.1 Statistical core and the similarity index

A set of weights is represented as a discrete probability distribution obtained from a histogram over the common range of the two compared sets. The core implements four distribution-similarity metrics: the **Kullback–Leibler (KL) divergence**, the **Jensen–Shannon (JS) divergence**, **cosine similarity** and the **Wasserstein (transport) distance**. KL and JS are computed in bits (base-2 logarithm).

The similarity index is a single scalar number that, at any moment of training, expresses how much the current weight distribution of the quantised model differs from the weight distribution of the reference model. For practical decision-making the JS divergence is key: it is metric and bounded in the interval $[0, 1]$, which makes it possible to set a fixed threshold above which the distribution is considered unacceptably distorted, and the values are directly comparable across datasets.

Crucially, the meaning of the index is context-dependent. In post-training quantisation across bit-widths, a higher divergence corresponds directly to greater degradation (fewer bits $\rightarrow$ larger distribution shift $\rightarrow$ lower accuracy). During quantisation-aware fine-tuning, however, the index typically grows together with accuracy, because the model adapts and its quantised weights legitimately move away from the original FP reference — in this regime the index measures adaptation, not degradation. The index is therefore always interpreted together with accuracy.

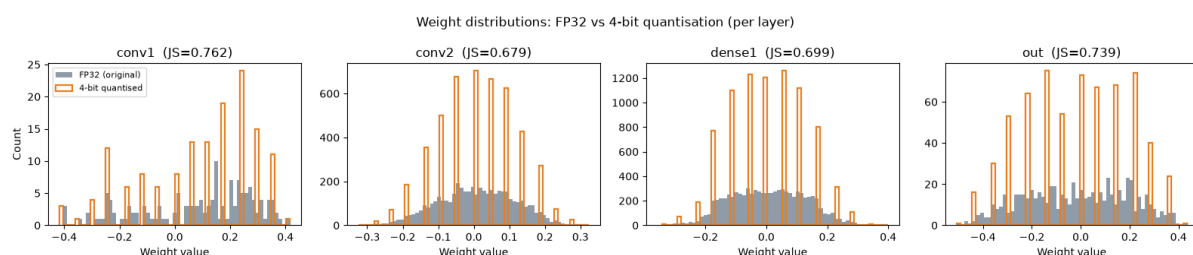


Fig. 1 — Visual comparison of the weight distribution before quantisation (FP32) and after it (4 bits), per layer. The continuous distribution breaks into narrow “spikes” corresponding to $2^4 = 16$ quantisation levels; it is precisely this change in shape that the similarity index quantifies.

2.2 Genuine quantisation-aware training (QAT)

For the training phase, post-training quantisation is not sufficient: if the model were simply fine-tuned in full precision after quantisation, its weights would return from the grid to arbitrary real values and the model would cease to be quantised. Genuine QAT solves this by having the **forward pass use quantised weights** (so the network learns to be robust to quantisation), while the **gradient is propagated through a straight-through estimator (STE)** — the rounding operation has a unit gradient in the backward pass, so the latent (full-precision) weights can be trained even though their quantised form is used in the forward pass. Custom quantised layers QuantConv2D and QuantDense are implemented, with **per-channel** quantisation (a separate scale for each output channel).

3. Implementation of the strategy

The core of the deliverable is a training strategy that integrates the statistical index into genuine QAT and uses it to control the process. It rests on three components that combine into a single procedure:

- **Monitoring callback** — after each epoch it extracts the effective quantised weights and computes their divergence against the frozen FP reference; it records the history of the

index (KL/JS/cosine) and the test accuracy, and optionally prints them live during training.

- **Index-driven early stopping** — terminates training once both the index and accuracy stabilise (the change in both drops below a threshold over a given number of consecutive epochs), saving computation without loss of quality.
- **Automatic bit-width selection** — using fast post-training quantisation it selects the lowest bit-width at which the global JS divergence does not exceed a set threshold, i.e. the most aggressive compression that still preserves acceptable distribution fidelity.

The complete procedure combines the components into a sequence: (1) index-driven bit-width selection → (2) building a QAT model initialised from the reference model → (3) genuine QAT with monitoring and index-driven early stopping. The loader is dataset-agnostic and supports sklearn digits, Fashion-MNIST, MNIST and CIFAR-10; the behaviour can be changed via configuration switches (e.g. CFG.run_cifar, CFG.run_imagenet, CFG.full_run).

4. Experimental results

4.1 Reference models

For each dataset, 5 independent reference (FP32) models were trained. Mean accuracy $\pm$ standard deviation: digits 0.9667 ± 0.0053 ; Fashion-MNIST 0.9073 ± 0.0041 ; MNIST 0.9875 ± 0.0014 ; CIFAR-10 0.6378 ± 0.0079 .

4.2 Verifying that quantisation is genuine

To demonstrate that this is not a pseudo-process, after a short QAT (4 bits) the number of **distinct values of the effective weights per channel** was counted. Since quantisation is per-channel, the correct criterion is the number of distinct values per channel, which must not exceed $2^b = 16$ — regardless of the number of epochs. The measured maxima (conv1: 9, conv2: 15, dense1: 15, out: 14) satisfy the criterion, confirming that the quantisation is genuine. For comparison, post-training quantisation of the reference model itself gives a global JS = 0.0002 at 8 bits (accuracy unchanged) and JS = 0.0967 at 4 bits.

4.3 Strategy vs. naive baseline

The strategy (SMT: automatic bit-width selection + monitoring + index-driven early stopping) was compared with a naive baseline (fixed 8-bit QAT for the same epoch budget, without monitoring and without auto-selection). With candidates of 4/6/8/16 bits, the strategy naturally selects approximately 6 bits (index well below threshold, accuracy preserved) — the analytically most economical lossless choice, with a compression of 5.33 $\times$ versus 4.0 $\times$ for the 8-bit baseline:

Dataset	FP32	SMT bits	SMT epochs	SMT acc.	SMT compr.	8-bit epochs	8-bit acc.	8-bit compr.
digits	0.9750	6	4	0.9722	5.33 $\times$	30	0.9778	4.0 $\times$
fashion_mnist	0.9091	6	4	0.9114	5.33 $\times$	30	0.9021	4.0 $\times$
mnist	0.9859	6	30	0.9906	5.33 $\times$	30	0.9924	4.0 $\times$
cifar10	0.6402	6	5	0.6494	5.33 $\times$	30	0.6295	4.0 $\times$

Table 1 — Comparison of the strategy (SMT, ~6-bit) with a fixed 8-bit baseline. SMT achieves comparable accuracy at higher compression, on several datasets with a substantially smaller number of epochs thanks to early stopping (e.g. digits 4 vs 30 epochs).

A note on deployment: although 6 bits is the analytically most economical choice, **8-bit (INT8)** remains relevant for deployment — integer hardware and runtimes (mobile and edge NPUs, INT8 accelerators, TFLite/ONNX) support 8 bits natively, whereas 6 bits is not.

4.4 Bit-width sensitivity

For each reference model, the immediate post-quantisation (PTQ) accuracy and the global JS divergence were measured across a range of bit-widths. The relationship **bit-width** → **divergence** is almost perfectly monotonic: Spearman's correlation coefficient $\rho = -0.984$ ($p = 7.5 \times 10^{-19}$) for all datasets — the index thus reliably tracks the severity of quantisation. For the relationship **divergence** → **accuracy** (pooled data, $N = 100$) $\rho = -0.452$ ($p = 2.3 \times 10^{-6}$), and divergence → loss $\rho = 0.403$.

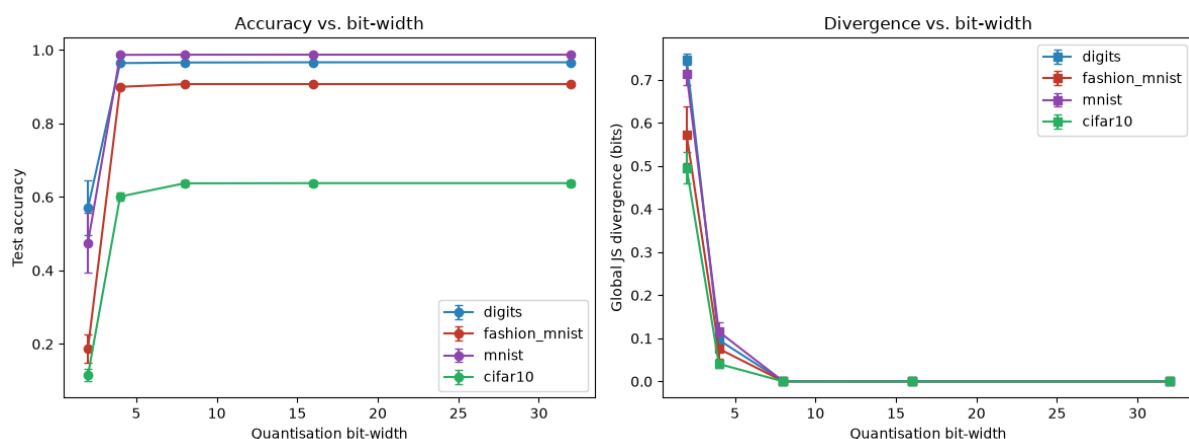


Fig. 2 — Accuracy (left) and global JS divergence (right) as a function of bit-width for four datasets. At high bit-widths the index is near zero and accuracy practically unchanged; below a certain limit the index rises sharply and a steep drop in accuracy follows.

4.5 QAT dynamics and computational overhead

During genuine QAT (4 bits), the evolution of the index and accuracy across epochs was tracked. Since the index is computed from the effective quantised weights, it captures the real adaptation of the model. The monitoring overhead was decomposed in order to isolate the cost of the divergence computation alone: **the divergence computation alone adds only +2.4%** to the training time; the costly part is only the optional per-epoch accuracy evaluation (+52%), which can be omitted in a production setting.

4.6 Scaling to ImageNet-V2

Generalisation to large models was carried out as post-training quantisation of several pretrained architectures (MobileNetV2, ResNet50) on a validation subset of ImageNet-V2 (~2,560 images). A quantisation-sensitive architecture (MobileNetV2) was deliberately mixed with a more robust one (ResNet50). At 8 bits the accuracy remains practically preserved (MobileNetV2 top-1 0.5193 vs FP32 0.5732; ResNet50 0.5617 vs 0.5631). At 4 bits ResNet50 is still usable (top-1 0.3646), whereas MobileNetV2 collapses (top-1 0.0018) — see the discussion of the limitations of the global index below.

4.7 Cross-dataset evaluation

Across all datasets — from small models to a large pretrained model on ImageNet — the same regularity appears: the index increases monotonically as the bit-width decreases, is near zero at high bit-widths with accuracy unchanged, and rises sharply below a certain limit, with the

steep drop in accuracy following only at even lower bit-widths. The index thus behaves as a dataset-agnostic early indicator of degradation — this is the main practical benefit of the strategy.

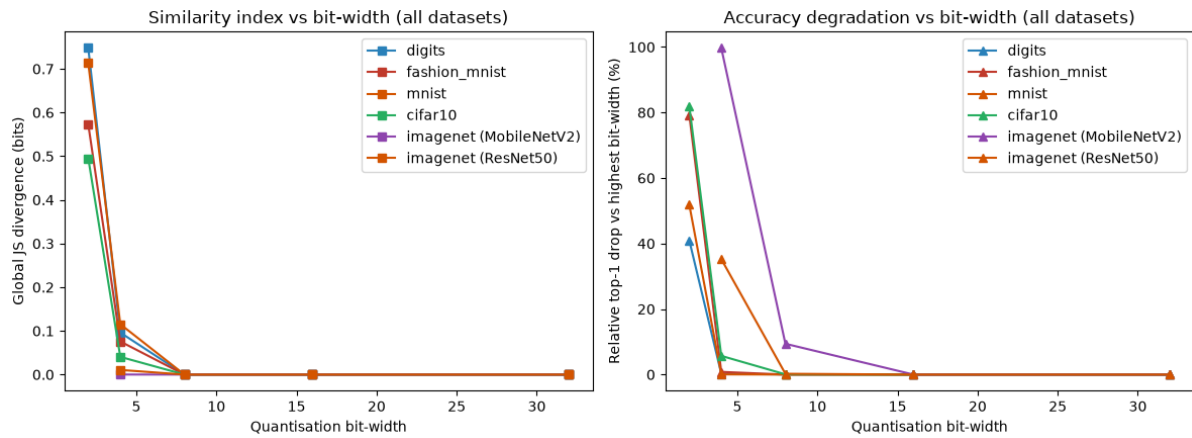


Fig. 3 — Similarity index (left) and relative top-1 accuracy drop (right) as a function of bit-width across all datasets. Since JS is bounded on $[0, 1]$, the values are directly comparable across datasets.

Values at the lowest tested bit-width:

Dataset (bit-width)	Index JS	Top-1 drop
digits (2 bits)	0.748	41.0%
fashion_mnist (2 bits)	0.573	79.3%
mnist (2 bits)	0.713	51.9%
cifar10 (2 bits)	0.495	81.9%
imagenet - MobileNetV2 (4 bits)	0.000	99.7%
imagenet - ResNet50 (4 bits)	0.011	35.2%

Table 2 — Index and accuracy drop at the lowest tested bit-width. Notable is the MobileNetV2 row (4 bits): the global JS remains near zero even though accuracy collapses — this case is addressed in the discussion.

5. Discussion and limitations

The main finding is that the global JS divergence of the weight distribution works as a **dataset-agnostic early indicator of degradation**: the monotonic relationship with bit-width ($\rho \approx -0.98$), the near-zero value when accuracy is preserved, and the threshold rise preceding the steep drop in accuracy make it possible to reliably select compression and control training.

An important limitation is visible in the MobileNetV2 architecture at 4 bits: the global JS of the weight distribution remains near zero, yet top-1 accuracy collapses (a drop of $\sim 99.7\%$). The global weight-distribution index is therefore a **necessary but not sufficient** condition for preserving accuracy — in architectures sensitive to quantisation (depthwise-separable convolutions, BatchNorm) the degradation may be concentrated in sensitive layers or in the activation distribution, which the global weight index does not capture. The index is therefore always interpreted together with accuracy, and per-layer and activation analysis (WP4) is needed for such cases.

A second limitation is the **reliability of early stopping**: it triggers reliably only where the index stabilises within the epoch budget during training (digits, MNIST); on other datasets (Fashion-MNIST) it may not trigger. It is therefore recommended to use it together with an epoch cap and an accuracy check, not on its own.

6. Conclusion and fulfilment of the deliverable

A new training strategy for quantised networks was implemented and experimentally validated, integrating the statistical weight-distribution index directly into genuine quantisation-aware training and using it to control the process. It was verified numerically that the effective weights remain genuinely quantised throughout the entire training (at most 2^b distinct values per channel). The strategy provides two actionable decisions — index-driven early stopping (a demonstrable saving of epochs while preserving accuracy) and automatic bit-width selection according to a divergence threshold — at negligible overhead for the index computation itself (+2.4%).

Fulfilment of deliverable D12. The deliverable fulfils its definition — the implementation of a new neural network training strategy based on a statistical method — with a functional, methodologically correct (genuine QAT), configurable and dataset-agnostic strategy, experimentally validated across multiple initialisations and datasets. All measured results (the full bit-width analysis, the evolution during QAT, per-layer sensitivity, summaries and ImageNet) are exported to CSV files, so the testing is auditable and reproducible. The total runtime of the reference experiment was ~4,720 s.

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